Jashore University of Science and Technology

Computer Science and Engineering

LitSphere

A Collaborative Systematic Literature Review, Benchmark, and Research Synthesis Platform

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Introduction & Research Motivation

CHALLENGE 01

Scientific Literature Explosion

Over 1.5 million academic papers are published annually. Researchers face severe cognitive overload attempting to manually extract, index, and synthesize critical evidence across hundreds of PDF manuscripts.

CHALLENGE 02

Fragmented Academic Toolchains

Reviewers are forced to juggle disconnected tools: Zotero for reference storage, Excel for data tables, and Word/Notepad for notes. This disjunction causes lost citations, data discrepancies, and workflow friction.

LIMITATION 03

Absence of Coordinate Traceability

Conventional spreadsheets store extracted claims with zero link to source text. Verifying a recorded metric requires re-reading multi-page papers from scratch due to missing bounding-box coordinate anchors.

THE SOLUTION

LitSphere Centralized Synthesis Platform

A unified, collaborative platform integrating split-screen PDF reading, coordinate-accurate claim anchoring, live synthesis matrix grids, and 1-click publication-ready LaTeX table generation.

Existing Systems & Limitations

REFERENCE MANAGERS

Zotero, EndNote

Strengths: Reliable bibliographic indexing, citation metadata generation, and local PDF file storage.
Critical Gaps: Incapable of structured variable extraction; zero comparative matrix synthesis; no PRISMA consensus screening.

MANUAL SPREADSHEETS

Microsoft Excel, Google Sheets

Strengths: Flexible rows and columns; universally accessible format.
Critical Gaps: Completely detached from PDF documents; severe data entry error rates; zero coordinate-anchored text jump; high formatting overhead.

SPECIALIZED SLR TOOLS

Rayyan, Covidence

Strengths: Structured title/abstract screening for medical reviews.
Critical Gaps: Prohibitive subscription paywalls; rigid clinical workflows; lack of interactive split-screen matrix extraction and benchmark clustering.

NOTE-TAKING SYSTEMS

Obsidian, Notion, LiquidText

Strengths: Freeform markdown notes and visual document linking.
Critical Gaps: Lack structured systematic review protocols; no multi-reviewer dual-blind voting; no automated LaTeX publication export.

LitSphere: Proposed System Architecture

CENTRALIZED COLLABORATIVE RESEARCH SYNTHESIS PLATFORM

Researcher Workspace: Dual-Pane Reviewer

- **Split-Screen PDF Reader:** Synchronous in-browser PDF viewing without external software.
- **Coordinate BBox Anchoring:** Highlights are permanently bound to PDF page coordinates.
- **Direct Variable Extraction:** Populate dataset, model, metrics, and limitations side-by-side.
- **Search & Tag Filter:** Instant filtering by keyword, publication year, and domain taxonomy.

Synthesis Lead: Master Matrix & Analytics

- **Interactive Master Matrix:** Unified comparative grid aggregating extracted variables across all papers.
- **Dynamic Clustering:** Automated thematic grouping by methodology, problem type, and results.
- **Dual-Blind PRISMA Engine:** Independent reviewer screening and Cohen's Kappa agreement check.
- **1-Click Academic Export:** Direct output to LaTeX `booktabs`, Excel (`.xlsx`), and BibTeX.

Core Impact: Eliminates the 80% administrative friction between reading academic literature and synthesizing publishable survey articles.

Feature Comparison: LitSphere vs. Existing Tools

Feature Dimension	Zotero	MS Excel / Sheets	Rayyan / Covidence	LitSphere (Proposed)
In-Browser PDF Annotation	✓ (Basic)	✗	✗	**✓ (Dual-Pane)**
Coordinate-Anchored Claims	✗	✗	✗	**✓ (Exact BBox)**
Master Synthesis Matrix Grid	✗	✓ (Manual)	✗	**✓ (Real-Time)**
Thematic Clustering	✗	✗	✗	**✓ (Taxonomy)**
Dual-Blind PRISMA Screening	✗	✗	✓ (Paid)	**✓ (Built-in)**
One-Click LaTeX `booktabs` Export	✗	✗	✗	**✓ (Native)**
Local-First SQLite WAL Storage	✓ (Proprietary)	✗	✗ (Cloud-only)	**✓ (Open ACID)**
Zero-Framework Sub-15ms Latency	✗	✗	✗	**✓ (Optimized)**

VERIFIABILITY

100% Traceability

Every extracted claim jumps directly to its original sentence in the PDF.

STANDARDS

PRISMA Compliance

Structured screening, consensus voting, and automated exclusion reporting.

EFFICIENCY

Publication-Ready

Direct generation of publication-grade LaTeX tables and BibTeX citation keys.

Advanced Features: LitSphere Advantage (Part 1)

FEATURE MODULE 01

Split-Screen Dual-Pane Reviewer

- Synchronous side-by-side layout: PDF reader docked alongside data extraction form.
- Eliminates disruptive Alt-Tab switching between viewer and external spreadsheets.
- Native PDF.js rendering with continuous vertical scrolling, zoom, and page jump.

FEATURE MODULE 02

Coordinate-Accurate Text Highlighting

- Bounding box coordinates recorded in normalized $[x, y, w, h]$ geometry.
- Highlighting survives zoom, scaling, and responsive viewport adjustments.
- Clicking an extracted matrix metric instantly navigates to the exact sentence.

FEATURE MODULE 03

Master Synthesis Matrix Grid

- Real-time multi-dimensional comparison table across all ingested papers.
- Inline editing with instant database persistence and audit trails.
- Dynamic filtering by publication year, taxonomy tags, and performance metrics.

Engineering Significance: These three interconnected capabilities create a closed-loop verification cycle, guaranteeing that every scientific claim in the synthesis matrix is verifiable in under 2 seconds.

Advanced Features: LitSphere Advantage (Part 2)

FEATURE MODULE 04

Dual-Blind PRISMA Screening

- Independent voting protocol (Include / Exclude / Maybe) for peer reviewers.
- Automated calculation of Cohen's Kappa (κ) inter-rater agreement score.
- Arbitration interface for lead supervisors to resolve inclusion conflicts.

FEATURE MODULE 05

Zero-Framework High Performance

- Built entirely using Vanilla ES6 JavaScript without bloated runtime frameworks.
- Sub-15ms client interaction latency and ultra-fast DOM redraw speeds.
- SQLite WAL mode enables concurrent reader execution with zero locking stalls.

FEATURE MODULE 06

1-Click Academic Export Engine

- Direct generation of IEEE/ACM compliant LaTeX `booktabs` code tables.
- Formatted Excel export (`.xlsx`) for statistical meta-analysis.
- Automatic BibTeX synchronization with sanitized citation keys and DOIs.

Research Rigor: Conforms to international PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) guidelines, ensuring standard-compliant literature reviews.

System Architecture & Technology Stack

FRONTEND LAYER

Native Web Architecture

- **HTML5 & CSS3:** Modular design tokens, light theme, CSS grid and flexbox.
- **Vanilla ES6 JS:** High performance, zero dependency bundle overhead.
- **PDF.js Library:** Mozilla client-side PDF rendering engine.
- **Canvas Overlay:** Coordinate highlight layer with precise BBox geometry.

BACKEND LAYER

Node.js REST API Service

- **Node.js & Express:** Non-blocking asynchronous event-driven I/O.
- **Multer Engine:** Streaming multipart PDF ingestion and validation.
- **JWT Security:** Stateless authorization with bcrypt hash encryption.
- **Export Pipeline:** Programmatic LaTeX table and XLSX synthesis.

PERSISTENCE LAYER

SQLite 3 Database Engine

- **SQLite 3 Engine:** Embedded, zero-configuration relational database.
- **WAL Mode:** Write-Ahead Logging for high-throughput concurrency.
- **ACID Guarantees:** Transactional integrity on all matrix cell writes.
- **JSON Extension:** Dynamic storage for extensible extraction schemas.

Design Philosophy: Local-first privacy, minimal deployment dependencies, maximum performance, and open standards compliance.

Expected Functional Modules

FOR RESEARCHERS

Extraction & Reading Tools

- Drag-and-drop batch PDF ingestion.
- Synchronized split-screen reader.
- Text selection and BBox anchor.
- BibTeX key and DOI lookup.
- Contextual annotation notes.
- Dynamic keyword cloud inspector.

FOR PROJECT LEADS

Synthesis & Governance

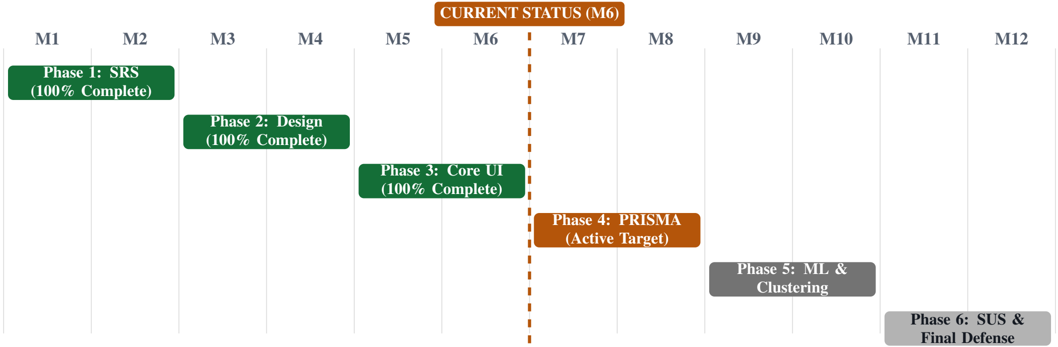
- PRISMA screening protocol manager.
- Extraction schema configuration.
- Real-time master matrix dashboard.
- Dynamic thematic clustering view.
- Reviewer conflict resolution tools.
- LaTeX `booktabs` & Excel export.

SYSTEM-WIDE

Core Infrastructure

- Role-Based Access Control (RBAC).
- Comprehensive action audit logging.
- Local-first offline data resilience.
- Automated database backups.
- Responsive cross-browser layout.
- REST API interface for extensions.

12-Month Project Timeline (Gantt Overview)



✓ **Months 1–6 (Complete):** Problem Analysis, Architecture, Reviewer & Matrix

➢ **Months 7–12 (Target):** PRISMA Consensus, Clustering, User Studies & Defense

Development Phases & Milestone Targets

Phase	Focus Domain	Timeline	Status	Key Deliverable
1	Problem Formulation & SRS	M1–M2	COMPLETED	Formal requirements specification document
2	Architecture & Schema Design	M3–M4	COMPLETED	Relational ERD, Use Case model, API contract
3	Core Reviewer & Matrix	M5–M6	COMPLETED	Split-screen PDF reader, BBox highlight, matrix grid
4	Collaborative PRISMA Screening	M7–M8	NEXT FOCUS	Dual-blind voting, Cohen's Kappa, conflict resolver
5	Synthesis Analytics & ML	M9–M10	PLANNED	Dynamic thematic clustering, keyword frequency
6	Benchmarking & Defense	M11–M12	PLANNED	SUS usability evaluation, thesis, final defense

MILESTONE 1: CURRENT STATUS

Current Deliverables (Month 6)

Complete functional system operational with PDF ingestion, coordinate-anchored annotations, interactive master matrix, and live LaTeX export.

MILESTONE 2: FUTURE TARGET

Final Defense Deliverables (Month 12)

Multi-user real-time PRISMA screening consensus engine, automated taxonomy clustering, empirical user study, and completed thesis dissertation.

SECTION 02

6-Month Progress Overview

Milestones Achieved, Core Deliverables, and Current System Status

Progress Summary: Month 1 to 6

Phase-Wise Completion Status

Phase	Completion
Phase 1: Literature Review & SRS	100%
Phase 2: Architecture & Database Design	100%
Phase 3: Reviewer & Master Matrix Grid	100%
Overall 12-Month Project Trajectory	55%

Key Metrics at Month 6

3 / 6
Phases Completed

55%
Overall Progress

12
Core Modules Active

Phase 4
Next Immediate Target

Executive Assessment: All preliminary research, system architecture, database modeling, and core functional implementation milestones for the first 6 months have been successfully executed according to the approved project plan.

Completed Modules (Month 1 to 6)

MODULE 01
User Auth & Access Control

Secure JWT session tokens, role-based route middleware, and bcrypt password encryption.

MODULE 02
Survey Initializer

Multi-project creation, research protocol configuration, and inclusion criteria setup.

MODULE 03
PDF Ingestion Pipeline

Drag-and-drop batch file ingestion, automatic BibTeX parsing, and metadata indexing.

MODULE 04
Dual-Pane PDF Reviewer

Synchronous in-browser PDF reader docked beside dynamic variable extraction forms.

MODULE 05
Coordinate BBox Highlighting

Geometric bounding box anchors bound to PDF coordinates for instant jump-to-quote.

MODULE 06
Master Matrix & Export

Live synthesis grid, inline cell editing, and 1-click LaTeX `booktabs` code output.

Remaining Work (Month 7 to 12)

MONTH 7–9: COLLABORATION & PRISMA

Consensus Screening Pipeline

- **Dual-Blind Screening Workflow:** Multi-reviewer independent vote recording without bias.
- **Cohen's Kappa (κ) Engine:** Automated inter-rater agreement computation and threshold alerts.
- **Supervisor Arbitration Panel:** Centralized dashboard for resolving reviewer inclusion discrepancies.
- **Automated Deduplication:** Title and DOI hash matching to eliminate redundant search imports.

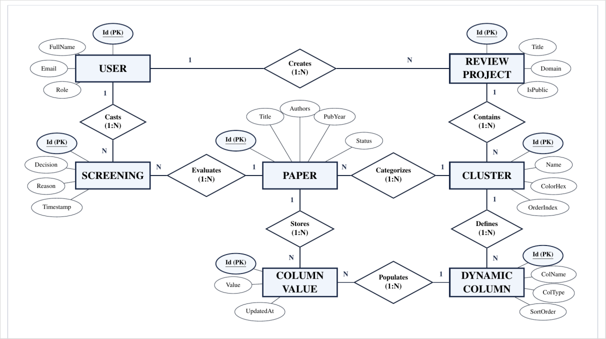
MONTH 10–12: ANALYTICS & FINAL DEFENSE

Synthesis Clustering & Usability

- **Dynamic Thematic Clustering:** Automated categorization by methodology, problem domain, and dataset.
- **User Evaluation (SUS):** Comprehensive usability study with academic research groups ($N \geq 20$).
- **Security Hardening:** SQL injection audit, rate limiting, and automated backup routines.
- **Dissertation & Defense:** Final thesis documentation and formal defense preparation.

Target Outcome: A production-ready, peer-reviewed collaborative systematic literature review software system.

System Design: Entity-Relationship Diagram



Relational Architecture

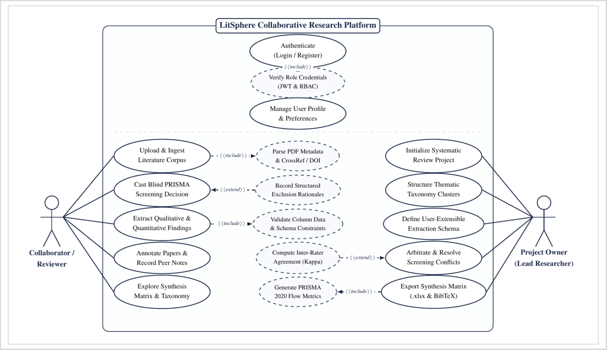
- **Core Entities:** `Users`, `Projects`, `Papers`, `Annotations`, `MatrixRows`, and `ExtractionFields`.
- **Relational Integrity:** Foreign keys with `ON DELETE CASCADE` prevent orphan records.
- **Coordinate Storage:** Precise bounding boxes stored as JSON-encoded geometric coordinates.
- **Concurrency Mode:** SQLite WAL enables high-frequency simultaneous reads and writes.

SCHEMA RESILIENCE

ACID Transactions

Guarantees atomic row updates across concurrent matrix edits.

System Design: Use Case Diagram



Actor & Role Workflows

- **Primary Actors:** Researcher (Reviewer), Project Lead (Supervisor), and System Administrator.
- **Researcher Actions:** Ingest papers, highlight text coordinates, extract synthesis variables, and view matrix.
- **Project Lead Actions:** Define research questions, configure extraction schemas, arbitrate conflicts, and export data.
- **Admin Actions:** Manage user accounts, role allocations, system audit logs, and backups.

ROLE GOVERNANCE

RBAC Security

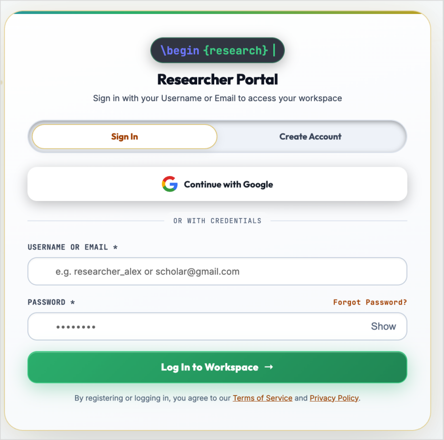
Enforces strict separation of concerns across review tiers.

SECTION 03

UI Implementation & Live Demo

Visual Inspection of the Fully Functional LitSphere System

UI: User Authentication & Session Login

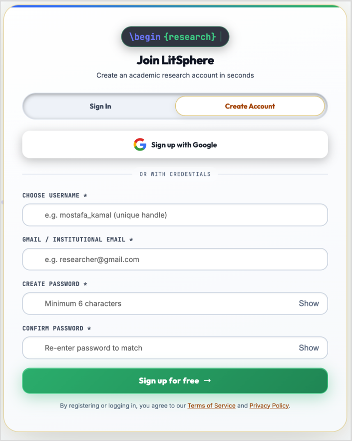


AUTHENTICATION LAYER

Single-Sign-On & Session Security

- **Regex Input Validation:** Instant client-side format checks for academic email addresses and password security.
- **JWT Bearer Generation:** Issues cryptographically signed stateless tokens upon successful credential verification.
- **Bcrypt Hashing:** High-entropy password encryption protecting user accounts from brute-force vectors.
- **Persistent Session State:** Secure local credential caching for seamless navigation across literature workspace tabs.

UI: Researcher Registration & Account Setup

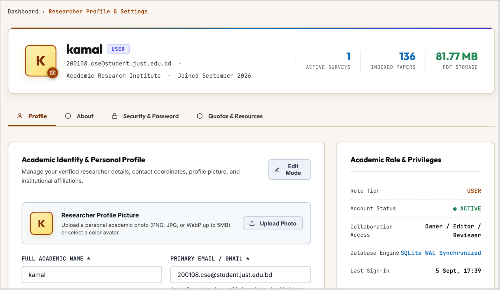


USER ONBOARDING

Institutional Registration & Role Assignment

- **Academic Profile Capture:** Records researcher full name, department, and university affiliation during signup.
- **Role-Based Governance:** Grants initial access tiers (Student Researcher, Reviewer, or Principal Investigator).
- **Password Policy Enforcement:** Multi-condition character validation ensures robust credential security.
- **Immediate Workspace Access:** Direct transition into systematic literature review projects without manual activation lag.

UI: Researcher Profile & Institutional Affiliation



ROLE GOVERNANCE

Account Credentials & Access Roles

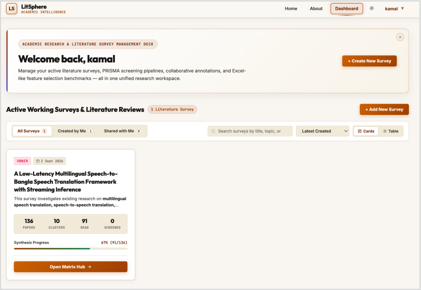
Displays active identity credentials, institutional department metadata, and user privileges across all assigned review teams.

ACTIVITY TELEMETRY

Project Memberships & Contributions

Monitors active literature survey projects, tracks individual paper extraction tallies, and manages password updates.

UI: Project Dashboard & Review Metrics



EXECUTIVE METRICS

Survey Portfolio & Extraction KPIs

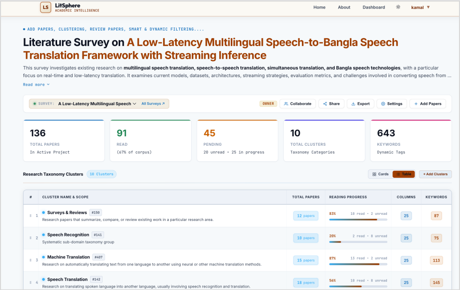
Real-time operational dashboard providing high-level progress tracking, paper ingestion counts, and variable extraction status.

WORKFLOW ACTIONS

Protocol Launch & Batch Ingestion

One-click shortcuts to initiate new PRISMA literature review protocols, import BibTeX bibliographic libraries, and add collaborators.

UI: Survey Workspace & Literature Ingestion



INGESTION PIPELINE

Drag-and-Drop Batch File Upload

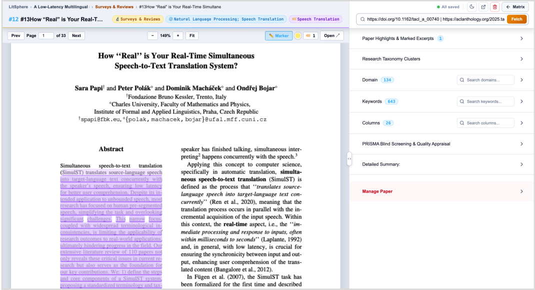
High-throughput multi-file PDF ingestion interface with automated bibliographic parsing, title hashing, and metadata normalization.

LITERATURE CATALOG

Status Filtering & Review Queue

Organized research repository featuring status pills (Uploaded, In Progress, Synthesized) and multi-field keyword filtering.

UI: Split-Screen Dual-Pane Paper Reviewer



DUAL-PANE WORKSPACE

Synchronous In-Browser PDF Reading

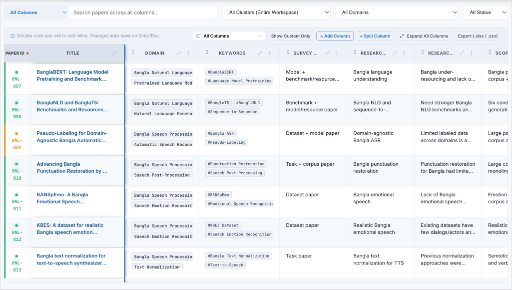
Integrated PDF reader docked side-by-side with extraction inputs, eliminating workflow disruption from external desktop viewers.

COORDINATE ANCHORING

Geometric BBox Text Highlighting

Text selections capture exact normalized bounding-box coordinates, providing direct jump-to-quote verification in under 2 seconds.

UI: Interactive Master Synthesis Matrix Grid



PAPER ID	TITLE	DOMAIN	KEYWORDS	SURVEY ...	RESEARC...	RESEARC...	SCOP
BNL-007	BanglaBERT: Language Model Pretraining and Benchmark...	Bangla Natural Language Pretrained Language Mod	#BanglaBERT #Language Model Pretraining	Model + benchmark/resource...	Bangla language understanding	Bangla under-resourcing and lack o...	Bangla p corpus +
BNL-008	BanglaNLG and BanglaT5: Benchmarks and Resources...	Bangla Natural Language Natural Language Genera	#BanglaT5 #BanglaNLG #Sequence-to-Sequence	Benchmark + model/resource paper	Bangla NLG and sequence-to-...	Need stronger Bangla NLG benchmarks an...	Six cond generati
BNL-009	Pseudo-Labeling for Domain-Agnostic Bangla Automatic...	Bangla Speech Processin Automatic Speech Recogn	#Bangla ASR #Pseudo-Labeling	Dataset + model paper	Domain-agnostic Bangla ASR	Limited labeled data across domains is a...	Large ps corpus a
BNL-010	Advancing Bangla Punctuation Restoration by ...	Bangla Speech Processin Speech Post-Processing	#Punctuation Restoration #Speech Post-Processing	Task + corpus paper	Bangla punctuation restoration	Punctuation restoration for Bangla had limite...	Large cc monolin
BNL-011	BANSpEmo: A Bangla Emotional Speech...	Bangla Speech Processin Speech Emotion Recognit	#BANSpEmo #Emotional Speech Recognitic	Dataset paper	Bangla emotional speech	Lack of Bangla emotional speech...	Emotion corpus a
BNL-012	KBES: A dataset for realistic Bangla speech emotion...	Bangla Speech Processin Speech Emotion Recognit	#KBES Dataset #Speech Emotion Recognition	Dataset paper	Realistic Bangla emotional speech	Existing datasets have few dialogs/actors an...	Realistic emotion
BNL-013	Bangla text normalization for text-to-speech synthesizer...	Bangla Speech Processin Text Normalization	#Bangla Text Normalization #Text-to-Speech	Task paper	Bangla text normalization for TTS	Previous normalization approaches were...	Semiotic and verb

CROSS-STUDY SYNTHESIS

Consolidated Comparative Grid

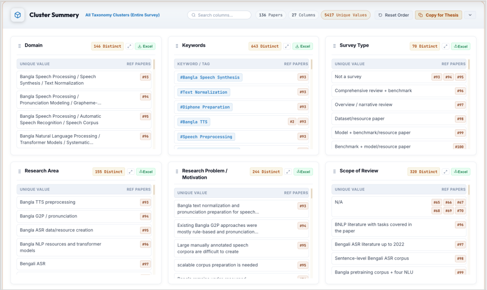
Multi-dimensional tabular matrix harmonizing extracted variables (methodology, datasets, metrics, limitations) across all papers.

TRANSACTIONAL EXPORT

Inline Editing & 1-Click LaTeX

Supports real-time inline updates with ACID database guarantees, plus direct 1-click export to publication-ready LaTeX booktabs code.

UI: Thematic Clustering & Synthesis Taxonomy



TAXONOMY DISCOVERY

Algorithmic Research Grouping

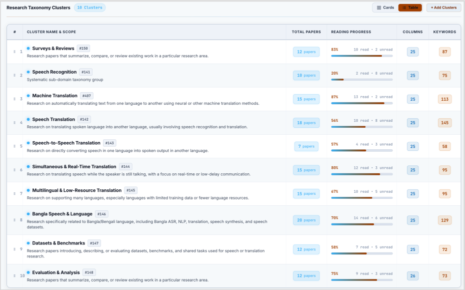
Automated thematic clustering categorizing ingested literature into distinct methodological paradigms, architectures, and focus domains.

SYNTHESIS ANALYTICS

Comparative Paradigm Distributions

Visual analytics displaying literature density, benchmark trends, and research frequency across surveyed publications.

UI: Benchmark Sample & Cluster Details



Research Taxonomy Clusters 10 Clusters

Cards Table + Add Clusters

#	CLUSTER NAME & SCOPE	TOTAL PAPERS	READING PROGRESS	COLUMNS	KEYWORDS
1	Surveys & Reviews #150 Research papers that summarize, compare, or review existing work in a particular research area.	12 papers	83% 10 read • 2 unread	25	87
2	Speech Recognition #141 Systematic sub-domain taxonomy group	10 papers	20% 2 read • 8 unread	25	75
3	Machine Translation #407 Research on automatically translating text from one language to another using neural or other machine translation methods.	15 papers	87% 13 read • 2 unread	25	113
4	Speech Translation #142 Research on translating spoken language into another language, usually involving speech recognition and translation.	18 papers	56% 10 read • 8 unread	25	145
5	Speech-to-Speech Translation #143 Research on directly converting speech in one language into spoken output in another language.	7 papers	57% 4 read • 3 unread	25	58
6	Simultaneous & Real-Time Translation #144 Research on translating speech while the speaker is still talking, with a focus on real-time or low-delay communication.	15 papers	80% 12 read • 3 unread	25	95
7	Multilingual & Low-Resource Translation #145 Research on supporting many languages, especially languages with limited training data or fewer language resources.	15 papers	67% 10 read • 5 unread	25	95
8	Bangla Speech & Language #146 Research specifically related to Bangla/Bengali language, including Bangla ASR, NLP, translation, speech synthesis, and speech datasets.	20 papers	70% 14 read • 6 unread	25	129
9	Datasets & Benchmarks #147 Research papers introducing, describing, or evaluating datasets, benchmarks, and shared tasks used for speech or translation research.	12 papers	58% 7 read • 5 unread	25	72
10	Evaluation & Analysis #148 Research papers that summarize, compare, or review existing work in a particular research area.	12 papers	75% 9 read • 3 unread	26	73

GRANULAR DRILLDOWN

Benchmark Sample Breakdown

Deep-dive inspection panel displaying constituent papers within a cluster, comparing evaluation metrics and algorithmic models.

RESEARCH GAP IDENTIFICATION

Cross-Methodology Discrepancies

Reveals performance discrepancies, benchmark dataset limitations, and unexplored research gaps across competitive methodologies.

Conclusion & Project Outlook

ACHIEVEMENTS

Phases 1–3 Complete

- End-to-end working system.
- Dual-pane PDF reviewer.
- Coordinate BBox anchoring.
- Live master matrix grid.
- Native LaTeX export.

STATUS

Current Milestone (M6)

- 55% roadmap executed.
- Zero major blockers.
- High UX performance.
- Proven architectural scale.
- On track for M12 defense.

NEXT STEPS

Months 7 to 12 Focus

- Collaborative PRISMA voting.
- Cohen's Kappa score engine.
- Supervisor conflict resolver.
- SUS usability evaluation.
- Final thesis dissertation.